# this in JavaScript contexts

Absolutely! Let's go through your entire code and expand it with **detailed explanations** and a **clear breakdown of how `this` behaves in each context in JavaScript** — covering:

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## 🔹 What is `this` in JavaScript?

- `this` refers to the **execution context**.
- The value of `this` **depends on how** a function is **called**, **not** where it is **defined**.
- It behaves **differently** in:
  - Global context
  - Regular functions
  - Methods inside objects
  - Arrow functions
  - Constructor functions / Classes
  - Event handlers
  - With or without `use strict`

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## 🔸 1. Global Context

### 🔹 In Browser:

```
js

console.log(this); // window
```

At the top level in a browser environment, `this` refers to the `window` object.

### 🔹 In Node.js:

```
js

console.log(this); // {}
```

In Node.js, `this` at the top level refers to the module's **exports object**, not the global object.

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## 🔸 2. Inside a Function

### 🔹 i. Non-Strict Mode:

```js
function greet() {
  console.log(this);
}
greet(); // window (in browser)
```

In non-strict mode, when a **regular function** is invoked, `this` refers to the **global object** (i.e., `window` in browsers).

### ii. Strict Mode:

```js
"use strict";
function meet() {
  console.log(this);
}
meet(); // undefined
```

In strict mode, calling a **regular function** results in `this` being `undefined`.

⚠️ **Important**: Without strict mode, JavaScript assigns `this` to `window`, which can lead to unexpected behavior.

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## 3. Inside a Method (Object Context)

```js
const obj = {
  name: "Rohit",
  age: 20,
  meet: function () {
    console.log(this.name);
  }
};

obj.meet(); // Rohit
```

Inside an object's method, `this` refers to the **object itself** (`obj` in this case).

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## 4. Arrow Functions and `this`

### Case 1: Arrow function directly in object

```js
let obj = {
  name: "rohit",
  greet: () => {
    console.log(this);
  }
};
```

```
obj.greet(); // Window or global object (NOT obj)
```

Arrow functions **don't have their own `this`**. They **lexically inherit `this`** from their surrounding scope. In this case, it's the **global scope**, not the **`obj`**.

### Case 2: Arrow function inside a method

```
js

let obj = {
  name: "rohit",
  greet: function () {
    let ab = () => {
      console.log(this.name);
    };
    ab();
  }
};

obj.greet(); // rohit
```

**`ab()`** is an arrow function **inside** a method. Arrow function inherits **`this`** from **`greet()`**, and **`greet()`** is called on **`obj`**, so **`this.name`** is **`"rohit"`**.

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## 5. Inside Constructor or Class

```
js

class Person {
  constructor(name, age) {
    this.name = name;
    this.age = age;
  }
}

let a = new Person("Rohit", 20);
console.log(a); // Person { name: "Rohit", age: 20 }
```

When you use **`new`**, **`this`** refers to the **newly created instance** of the class or constructor.

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## 6. Arrow Function in Global Scope

```
js

let greet = () => {
  console.log(this);
};
greet(); // window (browser)
```

Again, arrow functions don't have their own **`this`**. Here, it inherits from the **global context**.

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## 7. `this` in `setTimeout` and Callbacks

```
js

const obj = {
  name: "Rohit",
  greet: function () {
    setTimeout(function () {
      console.log(this.name); // undefined
    }, 1000);
  }
};

obj.greet();
```

Regular function inside `setTimeout` has its own `this`, which in non-strict mode is the `window`.

✅ **Fix using arrow function:**

```
js

const obj = {
  name: "Rohit",
  greet: function () {
    setTimeout(() => {
      console.log(this.name); // "Rohit"
    }, 1000);
  }
};
```

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## 🔸 8. `this` with `bind`, `call`, `apply`

### `call` and `apply` let you explicitly bind `this`:

```
js

function sayHello() {
  console.log(this.name);
}

let user = { name: "Aryan" };

sayHello.call(user); // Aryan
sayHello.apply(user); // Aryan
```

### `bind` returns a new function:

```
js

let boundFunc = sayHello.bind(user);
boundFunc(); // Aryan
```

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# ✅ Summary Table

| Context | `this` refers to |
| --- | --- |
| Global scope (browser) | `window` |

| Context | `this` refers to |
|---|---|
| Regular function (non-strict) | **`window`** |
| Regular function (strict) | **`undefined`** |
| Object method | The object itself |
| Arrow function | Lexical (surrounding scope) |
| Constructor function/class | The new instance being created |
| DOM Event Handler | The DOM element that fired the event |
| Using **`bind/call/apply`** | Whatever is explicitly passed |

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If you'd like, I can convert this into a **Markdown cheat sheet** or a **visual diagram**. Let me know!